

Tiny Mind Play School

A full-stack, enterprise-grade Next.js 16 preschool marketing platform and dual-portal management system with Supabase backend, RLS security, and real-time communication.

Version: 0.1.0 (Production Ready)

Target Node: v22+

Framework: Next.js 16.2 (App Router)

Database: PostgreSQL / Supabase

1 System Architecture & Core Philosophy

The platform is architected around a **hybrid portal model** catering to public visitors, school administrative staff, and student guardians with high performance, strict data isolation, and responsive UI.



Public Engagement Layer

Interactive web presence showcasing programs, philosophy, admissions, interactive 3D Dome Gallery, FAQs, and parent resources.



Admin Command Center

Complete control over student lifecycles, fee billing, automated receipt generation, daily attendance, notifications, and analytics.



Parent Self-Service Portal

Personalized dashboards for parents with real-time attendance logs, fee payment history, progress notes, and direct two-way messaging.



Dual Authentication Engine

Decoupled auth: Supabase Auth for staff + Edge-compatible JWT cookies for student admission-number logins.

2 Complete Technology Stack

LAYER	TECHNOLOGIES	KEY ROLE & IMPLEMENTATION
Frontend Core	Next.js 16.2.6 (App Router), React 19, TypeScript	Server and client components with strict typing across routes, UI, and stores
Styling & Design	Tailwind CSS 3.4, PostCSS, Radix UI Primitives	Custom design tokens, responsive cards, modals, and accessible dropdowns
Motion & 3D	Framer Motion 12.4, @use-gesture/react	Scroll reveals, micro-animations, and interactive 3D Dome Gallery with inertia physics
Database & Security	Supabase PostgreSQL + Row-Level Security (RLS)	Granular row-level permissions guaranteeing parents access strictly their child's records
Authentication	Supabase Auth + jose JWT + bcryptjs	Dual auth: Staff email/OAuth & Parent Admission ID with signed HTTP-only cookies
Data Interchange	xlsx (SheetJS), papaparse	Bulk student import/export and financial receipt generation
Testing & QA	Jest, React Testing Library, Playwright	Unit tests, component tests, and end-to-end multi-role browser test suites
Deployment & Edge	Vercel (bom1 Mumbai Region)	Edge proxy middleware, immutable static caching, and security headers

3 Authentication & Security Flow

Security is maintained through Next.js proxy middleware (`src/proxy.ts`) paired with Supabase RLS policies.

Auth Strategy: Staff log in using standard credentials authenticated against Supabase Auth. Parents log in via Admission Number + Password hash, issuing a tamper-proof signed JWT cookie (`parent_session`).

ROUTING PROXY (src/proxy.ts)

Request Path: `/dashboard/*`

1. Check "parent_session" Cookie (jose JWT)
 - └ Valid & mustChangePassword = true → Redirect `/auth/parent-change-password`
 - └ Valid parent role → Allow `/dashboard/parent/*` (Deny `/dashboard/admin`)
2. Check Supabase Auth Session (Admin / Staff)
 - └ Valid & role = 'admin' → Allow `/dashboard/admin/*` and `/dashboard/parent/*`
 - └ Unauthenticated → Redirect `/login` with return target

4 Detailed Portal Routes & Features

4.1 Public Website Routes

- `/` (**Home**): Hero dynamic carousel, trust pillars, curriculum cards, 3D gallery teaser, teacher profiles, and parent testimonials.
- `/about` : School legacy, pedagogy, mission, leadership profile, and campus highlights.
- `/programs` : Structured curricula (Infant Care, Toddlers, Playgroup, Nursery, Kindergarten, Daycare).
- `/gallery` : Full-screen 3D Dome Gallery enabling 360-degree interactive photo exploration.
- `/admissions` : Admissions roadmap, fee schedules, eligibility criteria, downloadable brochure, and online enquiry.
- `/contact` : Interactive map, operating hours, phone/email directory, and visitor dispatch form.
- `/parent-corner` : Child development resources, healthy eating guides, and handbook policies.
- `/login` : Unified portal switchable between Administrator and Parent admission logins.

4.2 Admin Management Dashboard (`/dashboard/admin/*`)

- `/dashboard/admin` : Operational KPIs (active enrollments, monthly collections, staff counts, attendance rates).
- `/dashboard/admin/students` : Student directory with class filters, search, modal editors, and Excel/CSV bulk import/export.
- `/dashboard/admin/fees` : Fee structure generation, partial/full payment logging, receipt generation, and status tracking.

- [/dashboard/admin/attendance](#) : Quick daily attendance marker across all programs with historical rollbacks.
- [/dashboard/admin/teachers](#) : Teacher and staff profiles, assigned classes, and contact directories.
- [/dashboard/admin/announcements](#) : Circular publisher with priority tags (Urgent, Event, General) and publishing toggle.
- [/dashboard/admin/enquiries](#) : Inbound prospective parent inquiry manager with lead statuses (New, Contacted, Enrolled).
- [/dashboard/admin/notes](#) : Student-specific feedback notes categorized by Academic, Behavioral, Health, or General.
- [/dashboard/admin/messages](#) : Real-time messaging inbox to chat directly with individual parents.
- [/dashboard/admin/reports](#) : Graphical analytics on enrollment trajectories, fee collection curves, and attendance.

4.3 Parent Portal (/dashboard/parent/*)

- </dashboard/parent> : Overview hub showing child's basic record, recent teacher remarks, upcoming events, and quick notices.
- </dashboard/parent/attendance> : Monthly interactive attendance calendar with color-coded Present, Absent, and Holiday badges.
- </dashboard/parent/fees> : Invoices, due dates, outstanding dues, payment history, and one-click printable PDF receipts.
- </dashboard/parent/announcements> : Filtered school circulars and notices relevant to the child's grade level.
- </dashboard/parent/events> : Activity calendar with event details, dates, and dress code instructions.
- </dashboard/parent/notes> : Historical remarks from teachers regarding academic growth and behavioral observations.
- </dashboard/parent/chat> : Direct real-time chat interface connecting parents directly to teachers and administrators.
- </dashboard/parent/profile> : Emergency contact records, address validation, and password reset panel.

5 Database Schema & Relationship Map

TABLE NAME	PRIMARY KEYS & REFERENCES	DESCRIPTION & RESPONSIBILITIES
students	id (PK) , admission_no (UQ)	Master records, admission credentials, password hashes, program level, parent info
profiles	id (PK → auth.users)	User identity tied to Supabase Auth; stores role ('admin' 'parent') and name
attendance	id (PK) , student_id (FK)	Daily attendance status (present , absent , late , excused) per date
fees	id (PK) , student_id (FK)	Billing records, fee types (Tuition, Transport, Activity), amount, due dates, payment status
payments	id (PK) , fee_id (FK)	Transaction records, auto-generated receipt numbers, payment methods, transaction timestamps
announcements	id (PK)	School-wide or class-specific circulars with priority flags and publish toggles
events	id (PK)	Academic calendar entries, festivals, parent-teacher meetings, holidays
notes	id (PK) , student_id (FK)	Teacher remarks and developmental milestone logs per student
teachers	id (PK)	Faculty directory with designations, bio, qualifications, and profile photos
enquiries	id (PK)	Leads generated from public admissions forms with follow-up tracking statuses
chat_messages	id (PK) , student_id (FK)	Two-way conversation history between school staff and parents

6 Codebase Directory Layout

```
preschool-website/
├─ public/                # Static imagery, badges, brand assets
├─ src/
│  ├─ app/                # Next.js App Router root
│  │  ├─ api/              # API Endpoints (chat, parent-login, enquiry, data-sync)
│  │  ├─ dashboard/
│  │  │  ├─ admin/        # 11 Dedicated Admin Management routes
│  │  │  └─ parent/       # 8 Parent Portal routes
│  │  ├─ (public pages)   # /, /about, /programs, /gallery, /admissions, /contact, /login
│  │  ├─ layout.tsx       # App wrapper (Font definitions, Header, Footer, Providers)
│  │  └─ globals.css      # Global styles & Tailwind layers
│  ├─ components/
│  │  ├─ DomeGallery/     # 3D interactive spherical gallery
│  │  ├─ chat/            # Chat window, message threads, unread badge counters
│  │  ├─ dashboard/       # DataTable, Modal, StatCard, EmptyState, Printable Receipt
│  │  └─ ui/              # Radix buttons, inputs, alerts, dropdowns
│  ├─ lib/
│  │  ├─ auth-context.tsx # Unified Auth Context provider
│  │  ├─ data-store.ts    # Central data layer with Supabase + fallback resilience
│  │  ├─ excel-import.ts  # Bulk student Excel & CSV parser
│  │  ├─ excel-export.ts  # Comprehensive reporting exporter
│  │  ├─ supabase.ts      # Browser Supabase client
│  │  └─ types.ts         # Canonical TypeScript domain types
│  └─ proxy.ts            # Next.js Edge proxy for route guarding
├─ schema.sql             # PostgreSQL DDL, Foreign Keys & RLS policies
├─ migration_*.sql        # Migration history scripts
└─ package.json           # Scripts, dependencies, and engine configs
```

7 Developer & Deployment Reference

Local Development & Build Commands

```
npm run dev  — Start local dev server at localhost:3000
npm run build — Build optimized production bundle
npm run test — Run Jest unit and component test suites
npm run test:e2e — Run Playwright browser automation suites
```

Environment Setup (`.env.local`)

```
NEXT_PUBLIC_SUPABASE_URL — Supabase project API URL
NEXT_PUBLIC_SUPABASE_ANON_KEY — Supabase public client key
SUPABASE_SERVICE_ROLE_KEY — Private server key for signing parent JWTs
```